

Shape-Gate Loss V2 Checkpoint Selection Pilot Campaign Results Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Executive Summary

The `shape_gate_loss_v2_checkpoint_selection_pilot_2026_07_21` campaign completed the approved one-run pilot on `polished_dataset` setpoints for the `Fw` surface.

The accepted campaign output directory is `output/training_campaigns/2026-07-21-13-03-55_shape_gate_loss_v2_checkpoint_selection_pilot_2026_07_21`.

The run completed without queue failures and produced the required campaign leaderboard, best-run YAML, best-run Markdown, metrics summary, training report, and checkpoint. It is accepted as a completed pilot, but it is not promoted to a full three-target, three-surface campaign on scalar evidence alone.

Pilot result

- run: `te_shape_gate_loss_v2_periodic_gru_sequence_fw_polished_setpoints` ;
- dataset: `polished_dataset` ;
- input mode: `setpoints` ;
- surface: `fw` / `forward_only` ;
- validation MAE: `0.001983` deg ;
- validation RMSE: `0.002445` deg ;
- test MAE: `0.001463` deg ;
- test RMSE: `0.001831` deg ;
- trainable parameters: `157,953` .

The scalar result improves materially over the first shape-gate loss pilot and beats the older same-backbone polished-setpoint `periodic_gru_sequence` `Fw` scalar baseline listed in the first pilot closeout. It remains a pilot-only checkpoint-selection result, because promotion still requires corrected shape-gated curve evidence and later full-matrix coverage.

Campaign Scope

The pilot intentionally used the narrowest approved scope:

Field	Value
Campaign	<code>shape_gate_loss_v2_checkpoint_selection_pilot_2026_07_21</code>
Dataset	<code>polished_dataset</code>
Input mode	<code>setpoints</code>
Dataset schema	<code>polished_setpoint_curve_v1</code>
Surface	<code>Fw</code>
Base model	<code>periodic_gru_sequence</code>
Loss profile	<code>shape_gate_v2_light_centered_offset_amplitude_harmonic</code>

Field	Value
Expected run count	1
Completed run count	1
Failed run count	0

The full promotion rule remains unchanged: any model family that survives a pilot must later be trained across `simplified_setpoints` , `polished_setpoints` , and `polished_actual_values` , with separate `global` , `Fw` , and `Bw` surfaces.

Remote Execution

The operator launched the campaign through the dedicated remote wrapper:

```
powershell
.\scripts\campaigns\cross_wave\run_shape_gate_loss_v2_checkpoint_selection_pilot_campaign.ps1 -Remote ""
```

The remote training itself completed successfully on `xilab-remote` . The original local SSH transport remained open before final marker capture, so the automatic sync-down stage did not complete. The completed remote artifacts were then recovered from `C:\Users\Martina Salami\Documents\Davide\Physics-Informed-Neural-Networks` and synchronized into this local repository.

Artifact Audit

Artifact	Path
Campaign leaderboard	<code>output/training_campaigns/2026-07-21-13-03-55_shape_gate_loss_v2_checkpoint_selection_pilot_2026_07_21/campaign_leaderboard.yaml</code>
Campaign best run	<code>output/training_campaigns/2026-07-21-13-03-55_shape_gate_loss_v2_checkpoint_selection_pilot_2026_07_21/campaign_best_run.yaml</code>
Campaign best-run report	<code>output/training_campaigns/2026-07-21-13-03-55_shape_gate_loss_v2_checkpoint_selection_pilot_2026_07_21/campaign_best_run.md</code>
Training metrics	<code>output/training_runs/shape_gate_loss_v2_checkpoint_selection_periodic_gru_sequence/2026-07-21-13-03-55_te_shape_gate_loss_v2_periodic_gru_sequence_fw_polished_setpoints/metrics_summary.yaml</code>
Training report	<code>output/training_runs/shape_gate_loss_v2_checkpoint_selection_periodic_gru_sequence/2026-07-21-13-03-55_te_shape_gate_loss_v2_periodic_gru_sequence_fw_polished_setpoints/training_test_report.md</code>
Best checkpoint	<code>output/training_runs/shape_gate_loss_v2_checkpoint_selection_periodic_gru_sequence/2026-07-21-13-03-55_te_shape_gate_loss_v2_periodic_gru_sequence_fw_polished_setpoints/checkpoints/periodic_gru_sequence-epoch=008-val_mae=0.00198279.ckpt</code>
Completed queue entry	<code>config/training/queue/shape_gate_loss_v2_pilot/completed/2026-07-21-13-03-55_001_001_shape_gate_loss_v2_periodic_gru_sequence_fw.yaml</code>
Family registry	<code>output/registries/families/shape_gate_loss_v2_checkpoint_selection_periodic_gru_sequence_fw/latest_family_best.yaml</code>

Metric Breakdown

Metric	Validation	Test
Total loss	0.021138	0.011095
MAE	0.001983	0.001463
RMSE	0.002445	0.001831
Pointwise loss	0.017484	0.009410
Centered curve-shape loss	0.014076	0.007082
Curve-offset loss	0.003407	0.002328
Curve-amplitude loss	0.048856	0.019544
Sparse harmonic-shape loss	0.000339	0.000146

Pilot Comparison

These are scalar training metrics, not official TE Curve Verification Pipeline decisions.

Family	Surface	Validation MAE	Test MAE	Decision
periodic_mlp_harmonic	global	0.001137	0.001270	Stronger scalar baseline
periodic_mlp_harmonic	Fw	0.001208	0.001442	Slightly stronger scalar baseline
shape_gate_loss_v2_checkpoint_selection_periodic_gru_sequence_fw	Fw	0.001983	0.001463	Completed v2 pilot; shape-gated screen required
periodic_gru_sequence	Fw	0.001832	0.002108	Older same-backbone scalar baseline; worse test MAE
wave4_1_mae_robust_loss	Fw	0.001792	0.002109	Worse test MAE than v2, still active shape-first road
wave4_2_quantile_p10_p50_p90	Fw	0.001801	0.002141	Worse test MAE than v2, still active shape-first road
shape_gate_loss_pilot_periodic_gru_sequence	Fw	0.002297	0.002522	First pilot; superseded by v2 scalar result

Acceptance Decision

The campaign is accepted as a completed pilot because:

- the one approved queue entry completed;
- the completed run uses `polished_dataset` setpoints and `Fw` as requested;
- the campaign leaderboard and best-run artifacts exist;
- the metrics summary, training report, checkpoint, and completed queue entry exist locally after remote recovery;
- the family registry and program registry were synchronized from the remote repository. The model is not accepted as a new active leader yet because:
- the pilot covers only one surface and one target;
- the corrected calibrated shape-gated reranker has not yet evaluated the checkpoint;
- the full promotion rule requires `simplified_setpoints`, `polished_setpoints`, and `polished_actual_values`, each across `global`, `Fw`, and `Bw`;
- official `TE Curve Verification Pipeline` decisions cannot be made from scalar `MAE` alone.

Recommended Next Step

Do not launch the full Aries matrix yet.

The next step should be a bounded checkpoint-level `TE Curve Verification Pipeline` screen for this v2 checkpoint on `polished_dataset` setpoints `Fw`, followed by the calibrated shape-gated reranker. If the v2 checkpoint keeps its scalar gain while improving shape, harmonic, derivative, and phase evidence against the first pilot and the reduced active `Fw` set, then prepare a separate full-matrix campaign package. If it fails the shape gate, stop this loss-profile branch and keep the shape metrics as validation and selection evidence rather than expanding the training objective.

Closeout Boundary

This closeout does not change the official `TE Curve Verification Pipeline` leaders and does not promote the v2 pilot to program baseline. The official selection boundary remains curve-first and per-surface.